

Canonical Jacobi–Wright–Fisher Diffusions at Every Positive Mutation Pair: Boundary Atlas, Complete Spectrum, Heat Determinant, and Moment Closure

Route-A structural certificate C200

Abstract

We close one canonical dynamical atlas for the Jacobi diffusion at every positive mutation pair. The source lock includes the conservative no-flux realization and a generator twice the common one-half Wright–Fisher clock. Scale and speed classify every endpoint, including equality; the Beta law makes the generator reversible; shifted Jacobi polynomials give the complete spectrum, sharp gap, heat kernel, trace-class determinant and all moments. The positive-recurrent sample process revisits neighborhoods, whereas a positive-time semigroup fixes only constant observables. Exact independent certificates test these distinctions but do not replace the continuum proof. The family has no intrinsic prime arithmetic or target determinant and is strictly rejected by Route A.

Keywords: Jacobi diffusion; Wright–Fisher process; Beta law; boundary classification; Jacobi polynomials; heat kernel; recurrence.

中文摘要

本文对全部正突变参数下的典范Jacobi–Wright–Fisher扩散给出完整图谱。源对象明确包含守恒零通量边界实现，且生成元时钟是常见二分之一约定的两倍。尺度与速度密度确定全部端点类型；Beta分布给出可逆性，Jacobi多项式给出完整谱、热核、迹类行列式与全部矩。样本路径正返并不矛盾于半群只有常值周期观测量。该系统没有内生素数算术，因此严格拒绝路线A。
关键词：Jacobi扩散；边界分类；可逆半群；谱隙；矩闭合。

1 Source lock and endpoint theorem

For $\alpha, \beta > 0$ let X be the conservative solution on $[0, 1]$ of

$$dX_t = [\alpha - (\alpha + \beta)X_t]dt + \sqrt{2X_t(1 - X_t)} dW_t, \quad (1)$$

with generator

$$Lf = x(1 - x)f'' + [\alpha - (\alpha + \beta)x]f'. \quad (2)$$

This is twice the clock in the neutral recurrent-mutation convention of Song and Steinrücken [2]; losing that factor halves every eigenvalue. When an endpoint is regular, the interior expression alone admits other extensions. We therefore freeze the canonical conservative/no-flux realization studied in the Wright–Fisher semigroup literature [1].

Theorem 1 (complete endpoint atlas). At zero the endpoint is regular reflecting for $0 < \alpha < 1$ and entrance for $\alpha \geq 1$. At one it is regular reflecting for $0 < \beta < 1$ and entrance for $\beta \geq 1$.

Proof. Up to positive constants the scale and speed densities are

$$s'(x) = x^{-\alpha}(1 - x)^{-\beta}, \quad m'(x) = x^{\alpha-1}(1 - x)^{\beta-1}.$$

The speed integral is finite at zero for every $\alpha > 0$, while the scale integral is finite precisely when $\alpha < 1$. Feller’s tests give regular versus entrance. The same calculation at one replaces α by β . The SDE/no-flux lock selects instantaneous reflection in the regular cases. In particular, equality belongs to the entrance side. $\square$

2 Beta reversibility and the complete spectrum

Normalize the speed law to

$$\pi(dx) = B(\alpha, \beta)^{-1} x^{\alpha-1} (1 - x)^{\beta-1} dx.$$

Direct differentiation gives the divergence identity

$$Lf = \pi^{-1} \frac{d}{dx} \{x(1 - x)\pi f'\}, \quad -\langle f, Lf \rangle_\pi = \int_0^1 x(1 - x) |f'|^2 d\pi. \quad (3)$$

Thus the canonical closure is reversible and self-adjoint. Define

$$J_n(x) = P_n^{(\beta-1, \alpha-1)}(2x - 1), \quad \lambda_n = n(n + \alpha + \beta - 1).$$

The Jacobi differential equation gives $LJ_n = -\lambda_n J_n$; orthogonal polynomial completeness gives an $L^2(\pi)$ basis [2, 3]. Therefore zero is simple, the sharp gap is $\lambda_1 = \alpha + \beta$, and

$$\|P_t f - \pi(f)\|_2 \leq e^{-(\alpha+\beta)t} \|f - \pi(f)\|_2.$$

Revision focus. Round 0 closes the realization, endpoint, reversibility and complete-spectrum theorem.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany the paper; no biological observations are used.

Ethics. No human, animal, personal, clinical or sensitive data are used; approval is not applicable.

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